

The Birth of UniForm: When Two Table Formats Finally Shook Hands 🤝

If you've worked with modern data lakes, you've probably heard people passionately debating **Delta Lake vs Apache Iceberg**—almost like Android vs iPhone.

But here's the funny part...

Both are trying to solve almost the same problem.

So why do we have both? And if Iceberg came first, why did Databricks bother creating Delta Lake?

Let's rewind the clock.

Act 1: Netflix had a headache 🍿

Around **2017**, Netflix was drowning in massive data lakes.

Parquet files were great at storing data, but they behaved like a pile of books thrown on the floor.

No ACID transactions.

No time travel.

No schema evolution.

No reliable updates or deletes.

Netflix needed a way to organize millions of data files that could be accessed by **multiple engines** like Spark, Hive, Presto (later Trino), etc.

So they built **Apache Iceberg**.

The philosophy was simple:

"One open table format that everyone can read."

Act 2: Databricks had a different headache ☕

Databricks looked at the same problem and said,

"That's cute... but our customers live inside Spark."

Their users wanted things like:

- UPDATE
- DELETE
- MERGE
- Streaming
- Lightning-fast Spark performance

Instead of optimizing for every engine, they optimized for **one engine**—Spark.

So in **2019**, they introduced **Delta Lake**.

Their philosophy was:

"Let's make Spark behave like a database on top of a data lake."

Same destination.

Different route.

Iceberg vs Delta: Rivals... but not really

One of the biggest misconceptions is thinking:

✗ Iceberg vs Parquet

The real comparison is:

Parquet → File Format

Delta Lake → Table Format

Apache Iceberg → Table Format



Here's the interesting part...

Both Delta Lake and Iceberg usually store their actual data as Parquet files!

The difference isn't the data.

The difference is **how they manage the metadata**.

Think of it like two librarians organizing the same library using different catalog systems.

Then something unexpected happened...

The industry grew up.

Companies no longer used just Spark.

Now they wanted:

- Spark
- BigQuery
- Snowflake
- Trino
- Flink
- Athena

...all reading the **same table**.

Iceberg's biggest strength suddenly became extremely valuable.

Meanwhile, Databricks customers started asking:

"Can my Delta tables also be read by other engines without copying the data?"

Awkward silence... 😅

Enter UniForm 🚀

Instead of forcing customers to choose between Delta and Iceberg, Databricks did something smart.

They introduced **UniForm**.

UniForm keeps:

- Delta metadata (for Spark & Databricks)
- Iceberg metadata (for external engines)

...while pointing both to the **same Parquet files**.

No duplicate data.

No expensive conversions.

Just two metadata "languages" describing the same table.

It's like writing the same book in English and Spanish so everyone can read it.

My biggest takeaway

I initially assumed Databricks created Delta because Iceberg didn't exist or wasn't capable enough.

That wasn't true.

Both companies were solving **different business problems**.

- Netflix optimized for **openness and multi-engine interoperability**.
- Databricks optimized for **deep Spark integration and performance**.

Fast forward a few years, and both ecosystems learned from each other.

Today, instead of fighting over which format wins, the industry is asking a much better question:

"How can every engine work on the same data without creating another copy?"

That's exactly the problem UniForm was born to solve.

Sometimes the biggest innovation isn't inventing something new... it's making two rivals finally speak the same language. 😊

#DataEngineering #Lakehouse #DeltaLake #ApacheIceberg #Databricks #BigData
#Spark #DataArchitecture #CloudComputing #AnalyticsEngineering